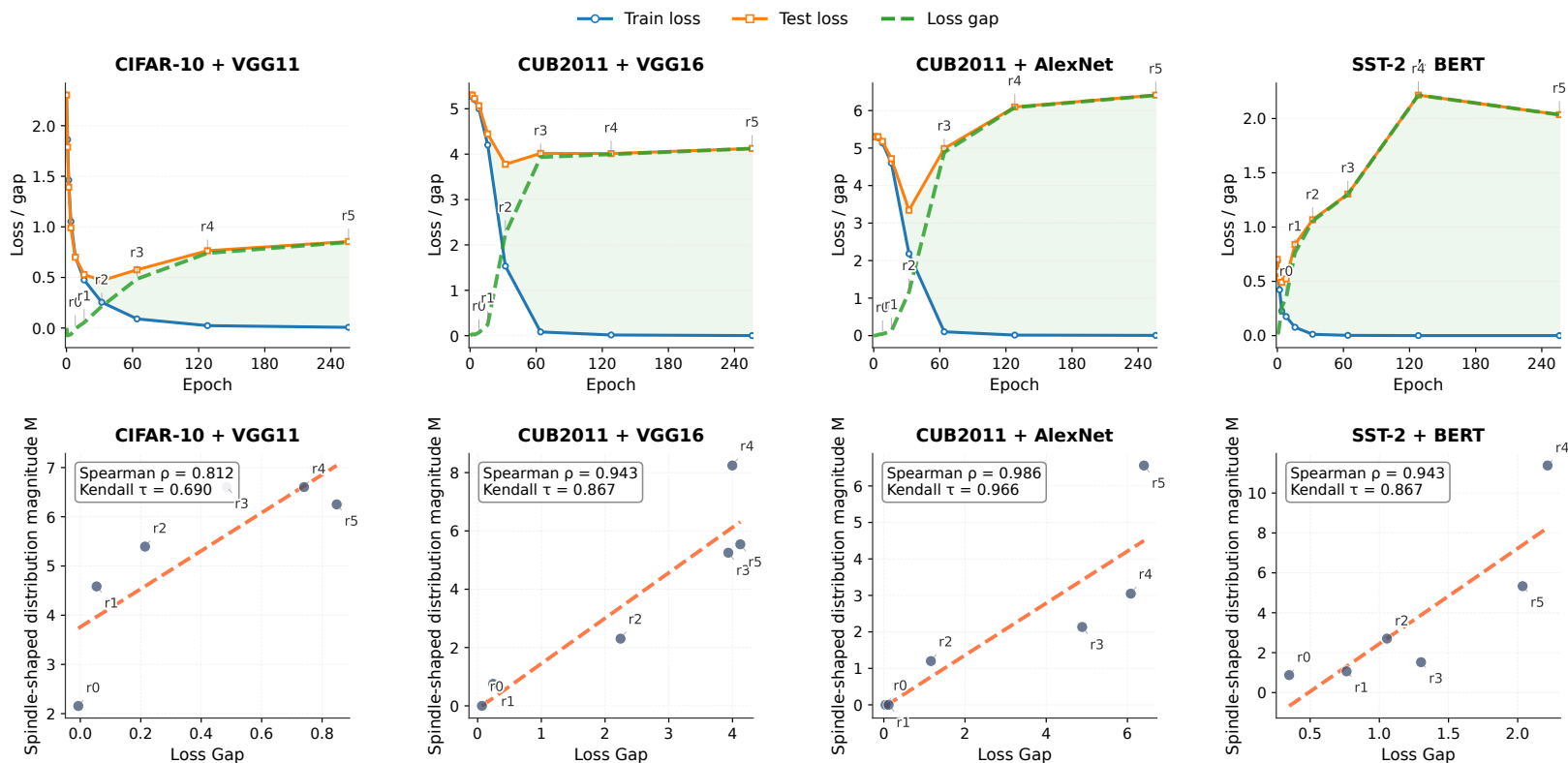




Caption. (Top) Train loss, test loss, and their gap (test loss – train loss) across epochs. (Bottom) Scatter plots of magnitude of spindled-shaped distribution M and the loss gap across different timepoints, with Spearman's ρ and Kendall's τ . High positive correlations indicate that a larger loss gap corresponds to a more pronounced spindle-shaped distribution, linking the spindle-shaped distribution of non-generalizable interaction to a the standard generalization metric, *e.g.*, loss gap.